RESULTS, CONCLUSION AND REFERENCES

7. Results and Key Findings

7.1 Summary of Results

The analysis of the supplied Formula 1 dataset produced the following findings:

1. The workbook contains 13,748 records across five worksheets, covering seasons from 2010 to 2026.
2. The race_results worksheet contains 6,464 records and 334 distinct race IDs.
3. A total of 11,464 pit stops are recorded, with a mean of approximately 1.77 per driver-race record.
4. The dataset contains 9,123 recorded DRS overtakes.
5. The lap_intelligence worksheet contains 3,814 records, including lap times, tyre information, and speed measurements.
6. Soft tyres have the highest number of recorded lap entries, at 1,202.
7. Dry is the most frequently recorded weather category, accounting for approximately 79.0% of distinct race IDs.
8. Max Verstappen has the highest cumulative points total in the supplied championship_standings worksheet, followed by Lewis Hamilton.

7.2 Value of the Visualizations

The Tableau visualizations make it easier to compare numerical values and understand categorical distributions.

Bar charts support comparisons between drivers, teams, and other categories. Pie charts help communicate proportions, while other visualizations can be used to examine performance measures and trends.

The dashboard brings selected charts together, providing a consolidated overview of the analysis.

7.3 Overall Project Outcome

The project demonstrates how a structured Excel dataset can be explored and transformed into meaningful visual summaries.

By combining exploratory data analysis, Tableau worksheets, a dashboard, a website deployed through Render, and an explanation video, the project presents the work through multiple formats.

The findings should be interpreted in the context of the supplied dataset, particularly where missing values or inconsistencies with official historical Formula 1 records exist.

8. Conclusion and Future Scope

8.1 Conclusion

The Formula 1 Racing Intelligence project demonstrates the use of data analysis and visualization to explore different aspects of Formula 1 racing.

The project involved examining a multi-sheet Excel workbook, identifying relevant variables, exploring statistical patterns, creating visualizations in Tableau, and assembling selected charts into a dashboard. A website was subsequently deployed using Render, and a video walkthrough was prepared to explain the work.

The analysis covered race results, qualifying performance, lap statistics, tyre compounds, pit stops, weather conditions, and championship standings.

The project highlights the importance of transforming raw data into meaningful visual information. Charts and dashboards help communicate statistical findings in an organized and accessible way.

Although the dataset contains missing values and some inconsistencies with official historical records, it provides a basis for demonstrating a data analytics and visualization workflow.

Overall, the project combines data exploration, statistical analysis, visualization, dashboard development, website deployment, and documentation into one integrated project.

8.2 Future Scope

The project can be improved in several ways:

1. Validate the dataset against reliable official sources.
2. Investigate missing values and inconsistent entries.
3. Develop additional visualizations to examine relationships between qualifying positions, lap times, and race results.
4. Add or improve dashboard filters for seasons, drivers, teams, and circuits where appropriate.
5. Examine changes in driver and team performance across seasons.
6. Explore relationships between pit-stop activity, qualifying performance, and race outcomes.

7. Improve the website's presentation, navigation, and accessibility.
8. Maintain detailed documentation of calculations, data sources, and visualization choices.

These improvements could make the project more reliable, informative, and useful for further analysis.

9. Project Links and References

9.1 Project Links

Live Website:
https://f1-race-intelligence-kp3a.onrender.com/

Tableau Dashboard or Workbook:
https://public.tableau.com/views/F1RaceIntelligenceDashboard_17930991864860/Dashboard1?:language=en-US&:display_count=n&:origin=viz_share_link

Project Explanation Video:
https://drive.google.com/file/d/1qN2UfdmmSYB0xsUczEuT15GT0aKuZFxR/view?usp=sharing

Dataset:
F1 Racing Project Dataset.xlsx — the Excel workbook provided for this project.

9.2 References

1. **Project Dataset:** *F1 Racing Project Dataset.xlsx*. Supplied Excel workbook containing race results, lap intelligence, qualifying results, championship standings, and circuit information.
2. **Tableau:** Official website and documentation for data visualization and dashboard development. https://www.tableau.com/
3. **Render:** Official website and documentation for website deployment. https://render.com/
4. **GitHub:** Official documentation for repositories, version control, and project documentation. https://docs.github.com/
5. **Google Docs:** Used to prepare and format the project report. https://docs.google.com/

The statistical findings presented in this report were calculated from the supplied project dataset. They describe the values contained in that workbook and have not been independently verified against official Formula 1 historical records.

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